CS 330 Final Project Design Decisions

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I modeled a desktop workspace with a monitor, keyboard, notebook, and mug. The objects use simple meshes because the reference scene depends more on their size and arrangement than on fine geometric detail. The monitor is the focal point: box meshes form its frame, screen, and base, while a tapered cylinder forms the support. A plane forms the desk beneath the objects. I placed the keyboard in front of the monitor, the notebook to the left, and the mug to the right, adjusting their X, Y, and Z positions so they sit on the desk without overlapping unintentionally.

I used a tiled wood image on the desk instead of stretching a single copy across the plane. The monitor uses separate textures for its frame, stand, and screen. I reused the dark plastic texture on the keyboard and notebook cover, then used simple material colors for the notebook pages and mug. The mug combines a cylinder with small boxes for its handle; a dark inset on top suggests its opening while keeping the geometry simple. Reusing the provided meshes and texture coordinates kept each object compact and made later placement changes easier.

Lighting was one of the parts I revised after the milestone feedback. A white point light above and in front provides the main illumination, and a purple point light adds a subtle accent. Two softer fill lights reach the side and rear surfaces that appeared dark in the previous version. Materials give the metal stand a stronger specular highlight than the wood and paper, while diffuse and ambient light keep the overall scene readable. I can adjust each light independently after checking the final render from several camera positions.

The camera accepts W and S for forward and backward movement, A and D for left and right, and Q and E for vertical movement. Mouse motion changes pitch and yaw without requiring a

position change. Scrolling adjusts movement speed within a bounded range, which helps when moving close to the small objects or across the wider desk. P selects perspective and O selects orthographic projection while the application runs. Both projections use the current camera orientation so changing the display does not unexpectedly reset the view.

The program separates reusable tasks into functions. SetTransformations builds each model matrix from scale, rotation, and translation; every object can call it with different values. LoadSceneTextures loads and binds the image files once, while SetShaderTexture selects a texture for a later draw call. SetTextureUVScale controls tiling independently for each mesh. DefineObjectMaterials stores named surface properties, and SetShaderMaterial applies those properties before an object is drawn. SetupSceneLights configures the lights in one place, leaving RenderScene focused on object placement and drawing. This separation would also help if I added more desktop objects later.